TSM 098 | INDEPENDENT PRACTICE

Polynomial operations and basic factoring

While TSM 099 studies Topic 17. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: like terms and distribution

1. Combine like terms.

$$(6x^2 + 2) + (-2x^2 - x - 2)$$

2. Combine like terms.

$$(5x^2 + 6x - 4) + (4x^2 + 2x + 5)$$

3. Combine like terms.

$$(-6x^2 - 5x - 2) + (5x^2 - 3)$$

4. Distribute and simplify.

$$5x(2x^2 + 4x - 2)$$

5. Distribute and simplify.

$$5x(x^2 + 4x - 5)$$

6. Distribute and simplify.

$$-3x(-x^2 - 3x - 5)$$

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Core: polynomial multiplication

7. Multiply and combine like terms.

$(x+4)(2x-2)$

8. Multiply and combine like terms.

$(x+4)(x-4)$

9. Multiply and combine like terms.

$(x-1)(x+3)$

10. Multiply and combine like terms.

$(x+4)(x+3)$

11. Multiply and combine like terms.

$(x^2+x+2)(x^2+2x+3)$

12. Multiply and combine like terms.

$(x^2+x-3)(x^2-x+1)$

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Core: introductory factoring

13. Factor completely; check by multiplying.

$x^2 + 7x + 12$

14. Factor completely; check by multiplying.

$x^2 - 8x + 12$

15. Factor completely; check by multiplying.

$x^2 + 2x - 15$

16. Factor completely; check by multiplying.

$x^2 - 6x - 16$

17. Factor completely; check by multiplying.

$6x^2 + 12x$

18. Factor completely; check by multiplying.

$x^2 - 16$

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Extension: mixed polynomial practice

19. Multiply and simplify.

$$(x^2+2x+1)(x^2+x-2)$$

20. Multiply and simplify.

$$(2x^2+x-3)(x^2-x+2)$$

21. Multiply and simplify.

$$(x+2)(x^2-4)$$

22. Factor completely and verify.

$$x^2-3x-28$$

23. Factor completely and verify.

$$x^2-10x+25$$

24. Explain the error: only the first and last terms were multiplied.

$$(x+2)(x+3)=x^2+6$$

TSM 098 | ANSWER KEY

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: like terms and distribution

1. Combine like terms.

$4x^2 - x$

2. Combine like terms.

$9x^2 + 8x + 1$

3. Combine like terms.

$-x^2 - 5x - 5$

4. Distribute and simplify.

$10x^3 + 20x^2 - 10x$

5. Distribute and simplify.

$5x^3 + 20x^2 - 25x$

6. Distribute and simplify.

$3x^3 + 9x^2 + 15x$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: polynomial multiplication

7. Multiply and combine like terms.

$2x^2 + 6x - 8$

8. Multiply and combine like terms.

$x^2 - 16$

9. Multiply and combine like terms.

$x^2 + 2x - 3$

10. Multiply and combine like terms.

$x^2 + 7x + 12$

11. Multiply and combine like terms.

$x^4 + 3x^3 + 7x^2 + 7x + 6$

12. Multiply and combine like terms.

$x^4 - 3x^2 + 4x - 3$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: introductory factoring

13. Factor completely; check by multiplying.

$(x+3)(x+4)$

14. Factor completely; check by multiplying.

$(x-2)(x-6)$

15. Factor completely; check by multiplying.

$(x+5)(x-3)$

16. Factor completely; check by multiplying.

$(x-8)(x+2)$

17. Factor completely; check by multiplying.

$6x(x+2)$

18. Factor completely; check by multiplying.

$(x-4)(x+4)$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: mixed polynomial practice

19. Multiply and simplify.

$$x^4 + 3x^3 + x^2 - 3x - 2$$

20. Multiply and simplify.

$$2x^4 - x^3 + 5x - 6$$

21. Multiply and simplify.

$$x^3 + 2x^2 - 4x - 8$$

22. Factor completely and verify.

$$(x + 4)(x - 7)$$

23. Factor completely and verify.

$$(x - 5)(x - 5)$$

24. Explain the error: only the first and last terms were multiplied.

$$(x + 2)(x + 3) = x^2 + 5x + 6$$

Every term in one factor must multiply every term in the other. The missing middle products are 3x and 2x.